

CROSS FLOW COOLING TOWERS

***“GLOBAL LEADER IN DESIGNING, MANUFACTURING AND
SUPPLY OF COOLING TOWERS AND ALLIED UTILITY SOLUTIONS***





"INNOVATING COOLING TECHNOLOGY FOR A SUSTAINABLE FUTURE"

BLUECHIP make crossflow cooling tower is a cooling system where water flows vertically downward through the fill media while air flows horizontally across it, creating a perpendicular heat exchange process. Hot water from process streams enters a distribution basin at the top of the tower, flowing downward through gravity, while air enters horizontally through louvers on the sides.

KEY FEATURES:

- **GRAVITY-BASED WATER DISTRIBUTION:** Eliminates the need for pressurized spray systems, simplifying maintenance.
- **AIRFLOW PATTERN:** Horizontal air movement across vertically falling water enables efficient heat transfer.
- **OPEN DESIGN:** Allows easier access for cleaning and maintenance.

ADVANTAGES:

- **ENERGY EFFICIENCY:** Gravity-fed water distribution reduces energy costs.
- **LOW MAINTENANCE:** Simplified design and easy access reduce operational expenses.
- **QUIET OPERATION:** Typically, quieter than counterflow designs.

CONSIDERATIONS:

Crossflow Cooling Towers require more space compared to counter flow designs and may be more susceptible to environmental contaminants due to their open design.

This Type of Cooling Tower is favored for its efficiency and low operational costs in environments where space and water cleanliness are manageable concerns



WHY CHOOSE A BLUECHIP MAKE CROSSFLOW COOLING TOWER?

Experience the ultimate in cooling tower technology with **BLUECHIP**'s Cross Flow Cooling Towers, engineered for high efficiency, durability, and seamless performance. Perfectly suited for industrial, commercial, and HVAC applications, our cooling towers combine advanced design and top-grade materials to deliver superior results.

KEY FEATURES:

- **OPTIMAL EFFICIENCY:** Engineered for effective thermal performance with reduced energy consumption.
- **RELIABLE DURABILITY:** Built with corrosion-resistant materials for long-term performance in demanding environments.
- **EASE OF MAINTENANCE:** Accessible design ensures straightforward servicing, reducing downtime and operational costs.
- **CUSTOMIZABLE DESIGN:** Tailored to meet your specific cooling requirements, with modular scalability.

TECHNICAL SPECIFICATIONS

Specification	Details
Cooling Capacity	Customizable (50 – 10000+ TR) per cell capacity
Flow Rate	50 m3/h to 2,000 m3/h per cell capacity
Material Options	Treated Timber, Pultruded FRP
Noise Levels	Low-noise operation (<85 dBA)
Water Drift Loss	<0.001%
Operating Temperature Range	Suitable for extreme conditions (3 Degrees to 25 Degrees)

COMPONENTS THAT BUILT A BLUECHIP CROSSFLOW COOLING TOWER

1. FILL MEDIA AND FILL SUPPORT GRIDS

- **FILL MEDIA:**

- **MATERIAL:** High-performance PVC or PP material for enhanced durability.
- **FUNCTION:** Maximizes the surface area for efficient heat exchange between water and air.
- **DESIGN:** Engineered to minimize clogging and ensure smooth operation.



- **FILL SUPPORT GRIDS:**

- **MATERIAL OF CONSTRUCTION (MOC):** Available in FRP (Fiberglass Reinforced Plastic) or SS304 (Stainless Steel 304).
- **PURPOSE:** Provides robust structural support for the fill media, ensuring stability and longevity.
- **CORROSION RESISTANCE:** Designed to withstand harsh operating conditions, including exposure to water and chemicals.

2. WATER DISTRIBUTION SYSTEM

- **GRAVITY-FED BASIN:** An open, gravity-fed distribution basin ensures water flow across the fill media.
- **NON-CLOG NOZZLES:** Strategically placed nozzles prevent clogging and ensure uniform water distribution.



3. DRIFT ELIMINATORS

- **PURPOSE:** Minimize water loss by achieving drift losses $>0.005\%$ reducing water losses
- **EFFICIENCY:** Captures water droplets while allowing maximum airflow.



4. AIR INLET LOUVERS AND EXTERNAL CASING

- **MATERIAL:** FRP / ACB
Corrosion-resistant louvers protect internal components.
- **FUNCTION:** Prevent debris entry and optimize airflow across the tower.

5. FANS AND MOTOR ASSEMBLY

- **FANS:** High-efficiency axial fans ensure smooth and quiet operation.
- **MOTORS:** Available in Direct Drive and Geared Driven systems for enhanced reliability and efficiency.

DIRECT DRIVE FAN MOTORS



- **ADVANTAGES:**
 - **REDUCED MECHANICAL COMPLEXITY:** Fewer moving parts compared to belt drive systems. Ideal for Small cooling towers
 - **HIGHER ENERGY EFFICIENCY:** Direct connection eliminates power losses from belts and gears.
 - **LOWER MAINTENANCE:** No belts to adjust or replace, elimination of gears and complex oil schedules, reducing downtime.

GEARED DRIVEN FAN MOTORS

- **ADVANTAGES:**

- **HIGH TORQUE CAPABILITY:** Suitable for larger cooling towers with heavy loads.
- **COMPACT DESIGN:** Efficient power transfer ensuring applicability in smaller footprint.
- **DURABLE AND ROBUST:** Designed for demanding industrial applications.



COMPARISON WITH BELT DRIVE SYSTEMS

- **EFFICIENCY:** Direct drive and geared systems deliver higher efficiency by reducing energy losses inherent in belt-driven systems.
- **MAINTENANCE:** Eliminates the need for frequent belt adjustments or replacements.
- **LONGEVITY:** Reduces wear and tear on components, extending the lifespan of the motor and fan assembly.

6. STRUCTURAL FRAME

- **MATERIAL:** Treated Timber, Heavy-duty Galvanized Steel, Stainless Steel, or Fiberglass for ultimate strength and corrosion resistance.
- **DESIGN:** Reinforced frame ensures structural stability, even in adverse conditions.



COOLING TOWER SAFETY FEATURES

- **FAN GUARD:** Protects against accidental contact with fan blades during operation.
- **HANDRAIL:** Provides additional safety for personnel working on the tower.
- **LADDER WITH EMERGENCY CAGE:** Ensures secure access to elevated areas with enhanced safety.
- **ANTI-SLIP WALKWAYS:** Provide safe access for maintenance personnel.
- **SAFETY GUARDS:** Protect fan openings and other moving components.
- **MONITORING INSTRUMENTS:** Equipped with an oil level switch and a vibration limit switch for real-time monitoring of critical parameters.



FAN GUARD



HANDRAIL AND LADDER
WITH SAFETY CAGE



INTERNAL WALKWAY WITH
HANDRAIL



OIL LEVEL SWITCH
AND VIBRATION
LIMIT SWITCH

COOLING TOWER ACCESSORIES

- **DOUBLE DECK STAIRCASE:** Provides safe and efficient access to multiple levels of the cooling tower.
- **HOT WATER BASIN COVER:** Prevents debris entry and reduces water evaporation.
- **DAVIT ARM:** Assists in lifting heavy components for maintenance or replacement, improving operational efficiency.



DOUBLEDECK STAIRCASE



FRP HOT WATER BASIN COVER



DAVIT ARM AND DOUBLEDECK STAIRCASE





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APPLICATIONS

BLUECHIP Counter Flow Cooling Towers are versatile and ideal for:

INDUSTRIAL APPLICATIONS

- Power plants
- Petrochemical and refining industries
- Steel and aluminum manufacturing
- Textile processing

COMMERCIAL APPLICATIONS

- HVAC systems in commercial buildings
- Data centers and IT infrastructure cooling
- Hospitals and healthcare facilities

SPECIALIZED INDUSTRIES

- Food and beverage manufacturing
- Pharmaceutical and chemical production
- Automotive and aerospace sectors

CUSMER SUPPORT AND SERVICES

- **CONSULTATION AND DESIGN:** Custom cooling solutions tailored to your specific needs.
- **INSTALLATION ASSISTANCE:** Expert installation and commissioning support.
- **MAINTENANCE PLANS:** Comprehensive service plans to ensure optimal performance.
- **24 / 7 TECHNICAL SUPPORT:** Our team is always ready to assist you.



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CONTACT US TODAY

Maximize your cooling efficiency and reduce operational costs with BLUECHIP's Counter Flow Cooling Towers. Let us help you achieve unparalleled performance for your industrial and commercial cooling needs.

PHONE: +91 9653150937 / 9321780765

EMAIL: office@bluechip-industries.com

WEBSITE: www.bluechipcoolingtower.com

FACTORY ADDRESS: Plot No - G88, MIDC Tarapur, MIDC Road, Behind Hotel RC, Boisar, Dist - Palghar, Maharashtra - 401506.

ADMIN OFFICE: 208 / 209 New Sonal Link Industrial Estate, Kanchpada, Malad West, Mumbai - 400064.





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LONG TERM ASSOCIATED CUSTOMERS & CONSULTANTS

Aarti Industries Ltd	Inox Air Products Ltd	Alfa Laval India
Deepak Nitrite Ltd	Coromandel International Ltd	Lupin India Ltd
Dorf Ketal India Ltd	Chemie Organics Ltd	UPL
Meghmani Organics Ltd	Sun Pharma Ltd	Zydus Lifesciences Ltd
IFFCO	Tata Steel India Ltd	Cipla Ltd
Pidilite India Ltd	Aditya Birla Group Ltd	Kumar Metal India
Servotech India	Valiant Organics Ltd	Anupam Rasayan Ltd
Deepak Phenolics Ltd	Aarti Drugs Ltd	Aarti Surfactants Ltd
Meghmani Finechem Ltd	Songwon India Ltd	ZCL Chemicals Ltd
Lubrizol India	Mitsuchem Plast Ltd	Hindustan Monomers Ltd
Hikal Ltd	Hetero India Ltd	Nath Industries Ltd
Institute of Chemical Technology (ICT), Mumbai	Associated Alcohol and Breweries Ltd (AABL)	Enmax Engineering India
Servotech India	Piramal Glass	Sajjan India Ltd
Kuwait Refinery	Mount Meru Millers	Alembic Pharma
Prasol Chemicals	Krishna Phoschem Ltd	Madhya Bharat Agro Ltd
Excelsource India	Time Technoplast Ltd	Chemie Organics Ltd
Indo Amines Ltd	Bluejet Healthcare Ltd	Echjay Industries Ltd

Toyo Engineering	Shroff Engineers	Lele Consultants
Technophil Consultants	TKT Consultants	Mott McDonald
L&T Engineering	Jacobs	JNCE Engineering
Enmax Engineering	Ketav Consultants	Universal MEP Projects and Engineering Services Limited
Zeppelin	Knexir	Aevitas



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PRESTEGIOUS PROJECTS

CUSTOMER NAME	SITE IMAGES
Aarti Industries Ltd.	
Aarti Drugs Ltd.	
Valiant Advance Science	
Calcimag Chemicals India	



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Alchemie Finchem Ltd.



Star Gases Pvt. Ltd.



Diamines and Chemicals Ltd



Nanda Chemicals Ltd





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Dynamic Products



Deepak Nitrite Ltd.



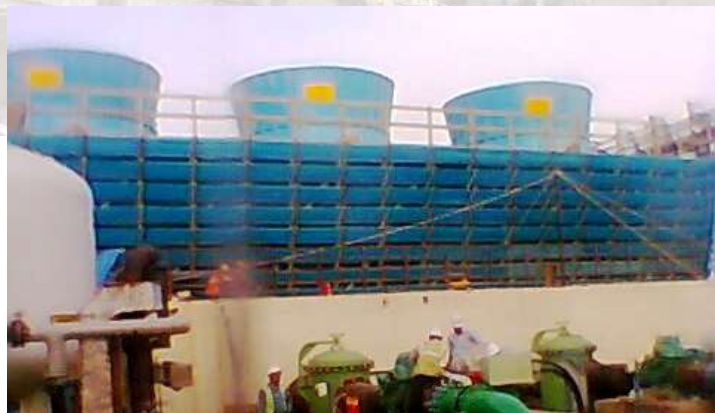
Kolhapur Oxygen



Prasol Chemicals



Inox Air Products



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